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INFUSOL® NAC
Acetylcysteine 200mg/mL Solution for Intravenous Infusion
Sterile and Non-Pyrogenic

Composition:

Each 25mL of solution contains 5g Acetylcysteine.

Product Description:

INFUSOL® NAC is a clear, colourless or very slightly pink-violaceous solution.

Pharmacodynamics/ Pharmacokinetics:

Pharmacology

N-acetylcysteine (NAC), active ingredient of INFUSOL® NAC, is a cysteine derivative provided with a free thiol (-SH) nucleophilic group which is able to interact directly with the electrophilic group of the oxidant radicals. Due to its molecular structure, NAC can easily cross the cellular membranes. Inside the cell, NAC is deacetylated to L-cysteine, an amino acid required for the synthesis of glutathione (GSH). GSH is a highly reactive tripeptide, found ubiquitously in various tissues of animals and it is essential for the maintenance of functional capacity as well as cellular morphological integrity, since it represents the most important protective endocellular mechanism against oxidant radical, either exogenous or endogenous, as well as towards numerous cytotoxic substances.

Due to its antioxidant property and its effectiveness in promoting the synthesis of glutathione, NAC is used as a substrate contributing to the cellular protection from harmful agent which, through progressive GSH depletion, would be able to express their cytotoxic action as in the case of paracetamol poisoning that may lead to hepatic failure, encephalopathy and death.

NAC administration is effective in the prevention of liver damage maintaining an adequate GSH level. The efficacy of the treatment is most likely to occur when it started early.

Toxicology

NAC has been evaluated to have low toxicity. After intravenous administration, the LD₅₀ is 2.8 g/kg in rat and 4.5g/kg in mouse. After oral administration, the LD₅₀ is more than 10 g/kg both in rat and mouse. During the repeated administration studies, the dose of 1 g/kg/die by oral route for 12 weeks has been well tolerated by rats. In dogs, no toxic reactions have been observed after oral administration of 300 mg/kg/die for one year. No teratogenic effect was observed in the offspring of pregnant rats and rabbits when treated with high dose of NAC during the organogenesis period.

Pharmacokinetics

The T_{max} was approximately 1 hour after oral administration. The bioavailability by the oral route was less than 10% of the dose administered. This finding does not mean that the drug is poorly absorbed but that extensive first-pass metabolism of the drug occurs in the liver. In plasma, NAC is found as N, N'-diacetylcysteine and mixed disulfides with other thiols or of low molecular weight or with proteins. The elimination half-life was 5-6 hours after intravenous administration. NAC is mainly excreted in the urine. The main urinary metabolite of NAC is the inorganic sulfate. However, in acute paracetamol intoxication, the urinary metabolite of NAC may differ.

Indication:

Antidote for paracetamol poisoning.

Recommended Dosage:

Diluted with Dextrose 5% and infused IV. Initial, 150 mg/kg IV in 200 ml over 60 minutes, then 50 mg/kg IV in 500 mL over 4 hours, followed by 100 mg/kg IV in 1000 mL over 16 hours. Total dose: 300mg/kg in 20 hours. The reconstituted solution is stable for 24 hours at a temperature not exceeding 30°C.

Route of administration: Intravenous (I.V)

Contraindications:

Known hypersensitivity to the drug or to other chemically related substances. Contraindicated in children under 2 years of age.

Warnings and Precautions:

Hypersensitivity Reactions

Serious acute hypersensitivity reactions during NAC administration including rash, hypotension, wheezing, and/or shortness of breath, have been observed in patients receiving intravenous NAC for paracetamol overdose and occurred soon after initiation of the infusion. If a severe hypersensitivity reaction occurs, immediately stop the infusion of NAC and initiate appropriate treatment.

Acute flushing and erythema of the skin may occur in patients receiving NAC intravenously. These reactions usually occur 15 to 60 minutes after initiating the infusion and often resolve spontaneously despite continued infusion of NAC. If a reaction to NAC involves more than simply flushing and erythema of the skin, it should be treated as a hypersensitivity reaction.

Management of less severe hypersensitivity reactions should be based upon the severity of the reaction and include temporary interruption of the infusion and/or administration of antihistaminic drugs. The NAC infusion may be carefully restarted after treatment of the hypersensitivity symptoms has been initiated; however, if the hypersensitivity reaction returns upon re-initiation of treatment or increases in severity, NAC should be discontinued and alternative patient management should be considered.

Patients suffering from bronchial asthma or with a history of asthma, must be closely monitored since bronchospasm may occur during treatment. In such event, the treatment must be suspended immediately. It should be used with caution in asthmatic patients and in patients with a history of peptic ulceration. Intravenous administration must be done in a hospital under strict medical surveillance. The administration of NAC at high doses may reduce the prothrombin time. However, it is not established if this is a result of the biologic action of NAC. In any way, the monitoring of coagulation factors is advisable particularly in cases of liver transplant. NAC may interfere with the determination of salicylates (colorimetric method) and with the determination of plasma and urinary ketones (nitroprusside test). The possible presence of sulfur-like odor does not indicate an alteration of the product but is characteristic of the active ingredient contained in this preparation.

Interactions with Other Medicaments:

Interactions between acetylcysteine and antibiotics have been reported. However, these are not remarkable when NAC is administered as treatment for paracetamol poisoning. Mixing INFUSOL® NAC with other drugs is not advised. Since NAC can react chemically with rubber, iron, copper etc., it is advisable to use glass or plastic material.

Statement on usage during pregnancy and lactation:

Animal studies indicated no evidence of teratogenic effect of NAC. The decision to use NAC at antidotal doses during pregnancy and lactation should be based on assessment of the risk-to-benefit ratio taking into consideration the peculiar characteristics of each case and in particular, the possible risk to the mother and foetus induced by the offending substance in the absence of specific treatment. In any case, lactation must be suspended in case of poisoning.

Adverse Effects/ Undesirable Effects:

After intravenous administration of NAC, the following anaphylactic reactions have been rarely reported: urticaria, angioedema, bronchospasm, nausea, vomiting, hypertension, tachycardia, dizziness and fever. These are probably pseudo-allergic reactions due to the histamine release. These symptoms may require an urgent therapy.

Immune System Disorders:

Anaphylactic/anaphylactoid reaction

Skin and Subcutaneous Tissue Disorders:

Severe cutaneous adverse reactions (SCAR) e.g. erythema multiforme, Stevens-Johnson syndrome (SJS) and toxic epidermal necrolysis (TEN). In most of these cases reported at least one other drug was administered at the same time, which may have possibly enhanced the described mucocutaneous effects.

Overdosage and Treatment:

Cases of overdose were characterised by similar symptomatology than those observed in cases of adverse events. In case of overdose, treatment must be suspended. Management of overdose is based on symptomatic treatment and/or resuscitation.

Incompatibilities:

None known.

Storage Condition Before and After Dilution:

Do not store above 30°C.

Shelf Life:

3 years from manufacturing date in the proposed storage condition. The solution should be used within 24 hours after dilution. Do not use after expiry.

Dosage forms and packaging available:

25 mL x 20 glass vials per box

Manufacturer / Product Registration Holder:

AIN MEDICARE SDN. BHD.

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